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Engaging module

Abstract

An engaging module has a male engaging element and a female engaging element. The male engaging element is connected to the female engaging element and has a first casing, a first magnetic element disposed in the first casing, and multiple matching structures disposed around an external surface of the first casing. Each matching structure has a guiding notch, an extending channel, and a positioning channel. The female engaging element has a second casing, a second magnetic element disposed in the second casing, and multiple positioning protrusions respectively corresponding to the multiple matching structures. The positioning protrusions respectively engage in the positioning channels when the first magnetic element and the second magnetic element are magnetically attracted to each other, and will not leave the positioning channels when an external pulling force is applied on the engaging module, and this simultaneously improves the convenience of assembly and the stability after assembly.

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Background/Summary

BACKGROUND OF THE INVENTION

1. Field of the Invention

(1) The present invention relates to an engaging module, and more particularly to an engaging module that can enable two parts of an object to assemble or disassemble.

2. Description of Related Art

(2) When connecting or assembling two opposite ends of a chain object, an engaging module is disposed between the two opposite ends of the chain object. The engaging module has a screwed element and an assembling element. The screwed element has an external thread, and the assembling element has a screwed hole. The screwed element and the assembling element are respectively disposed on the two opposite ends of the object, and the two opposite ends can be disassembled or assembled with each other according to a user's need via the external thread of the screwed element screwed with the screwed hole of the assembling element.

(3) However, when the user connects the two opposite ends of the object, the user needs to align the screwed element with the assembling element to screw the screwed element into the assembling element to connect the two opposite ends of the object, and this is inconvenient for the user to align

and assemble the screwed element and the assembling element to connect the two opposite ends of the object.
(4) In order to make the engaging module more convenient in use, at present, a conventional magnetic engaging module has been provided on the market, and the conventional magnetic engaging module has a male magnetic buckle and a female magnetic buckle. The male magnetic buckle and the female magnetic buckle are respectively provided with magnetic units that can attract each other. When the user operates the conventional magnetic engaging module, the user can directly move the male magnetic buckle close to the female magnetic buckle, and the magnetic units in the male magnetic buckle and the female magnetic buckle can be attracted to each other, thereby achieving the purpose of engaging, and this can effectively improve the convenience of alignment and assembly. Although the conventional magnetic engaging module can effectively improve the convenience of alignment and assembly, the male magnetic buckle and the female magnetic buckle of the conventional magnetic engaging module are prone to separate from each other via an external force, so the stability of the conventional magnetic engaging module after assembly is still insufficient.
(5) To overcome the shortcomings, the present invention provides an engaging module to mitigate or obviate the aforementioned problems.
SUMMARY OF THE INVENTION
(6) The main objective of the present invention is to provide an engaging module that can enable two parts of an object to assemble or disassemble.
(7) The engaging module in accordance with the present invention has a male engaging element and a female engaging element. The male engaging element is connected to the female engaging element and has a first casing, a first magnetic element disposed in the first casing, and multiple matching structures disposed around an external surface of the first casing. Each matching structure has a guiding notch, an extending channel, and a positioning channel. The female engaging element has a second casing, a second magnetic element disposed in the second casing, and multiple positioning protrusions respectively corresponding to the multiple matching structures. The positioning protrusions respectively engage in the positioning channels when the first magnetic element and the second magnetic element are magnetically attracted to each other, and will not leave from the positioning channels when an external pulling force is applied on the engaging module, and this simultaneously improves the convenience of assembly and the stability after assembly.
(8) Other objectives, advantages and novel features of the invention will become more apparent from the following detailed description when taken in conjunction with the accompanying drawings.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
(1) FIG. **1** is a perspective view of an engaging module in accordance with the present invention;
(2) FIG. **2** is an exploded perspective view of the engaging module in FIG. **1**;
(3) FIG. **3** is an enlarged side view in partial section of the engaging module in FIG. **1**;
(4) FIGS. **4** and **5** are operational side views in partial sections of the engaging module in FIG. **1**; and
(5) FIG. **6** is an operational side view in partial section of the engaging module in FIG. **1**, pulled by an external force.
DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENT
(6) With reference to FIGS. **1** to **3**, an engaging module in accordance with the present invention comprises a male engaging element **10** and a female engaging element **20**. The male engaging

element **10** is assembled with the female engaging element **20** along an assembling axis L.

(7) With reference to FIGS. **1** to **3**, the male engaging element **10** has a first casing **11**, a first magnetic element **12**, and multiple matching structures **13**. The first casing **11** has two ends, an assembling end **111**, an assembling hole **112**, a connecting end **113**, and a first through hole **114**. The assembling end **111** is formed on one of the two ends of the first casing **11**. The assembling hole **112** is formed in the assembling end **111** of the first casing **11**. The connecting end **113** is formed on the other one of the two ends of the first casing **11**. The first through hole **114** is formed through the connecting end **113** of the first casing **11**. The first magnetic element **12** is disposed in the assembling hole **112** of the first casing **11**.

(8) With reference to FIGS. **1** to **3**, the multiple matching structures **13** are disposed around an external surface of the first casing **11**, and each one of the multiple matching structures **13** has a guiding notch **131**, an extending channel **132**, a positioning channel **133**, a positioning area **134**, a limiting area **135**, a limiting protrusion **36**, and a non-return protrusion **137**. The guiding notch **131** may be V-shaped and is formed on the external surface of the first casing **11** adjacent to the assembling end **111** of the first casing **11**. The extending channel **132** is formed on the external surface of the first casing **11**, communicates with the guiding notch **131**, and is obliquely extended toward the connecting end **113** of the first casing **11**. The positioning channel **133** is formed on the external surface of the first casing **11**, is connected to and communicates with the extending channel **132** opposite to the guiding notch **131**, and has two ends. One of the two ends of the positioning channel **133** extends toward the connecting end **113** of the first casing **11**, and the other one of the two ends of the positioning channel **133** extends toward the assembling end **111** of the first casing **11**.

(9) With reference to FIGS. **2** and **3**, the positioning area **134** is formed on the end of the positioning channel **133** that extends toward the connecting end **113** of the first casing **11**. The limiting area **135** is formed on the end of the positioning channel **133** that extends toward the assembling end **111** of the first casing **11**, and communicates with the positioning area **134**. The guiding notches **131** of the multiple matching structures **13** are connected to and communicate with each other. The limiting protrusion **136** is formed on and protrudes from the external surface of the first casing **11** between the limiting area **135** and the extending channel **132**. The non-return protrusion **137** is formed on and protrudes from the external surface of the first casing **11** between the positioning area **134** and the extending channel **132**. Furthermore, an imaginary connecting line D is formed between the limiting protrusion **136** and the non-return protrusion **137** and is non-parallel to the assembling axis L.

(10) With reference to FIGS. **1** to **3**, the female engaging element **20** has a second casing **21** and a second magnetic element **22**. The second casing **21** has two ends, a mounting end **211**, a mounting hole **212**, multiple positioning protrusions **213**, a linking end **214**, and a second through hole **215**. The mounting end **211** is formed on one of the two ends of the second casing **21**. The mounting hole **212** is formed in the mounting end **211** of the second casing **21**, and the assembling end **111** of the first casing **11** extends into the second casing **21** via the mounting hole **212**. The multiple positioning protrusions **213** are formed on and protrude from an internal surface of the second casing **21** and respectively correspond to the multiple matching structures **13** of the male engaging element **10**. The second magnetic element **22** is disposed in the mounting hole **212** of the second casing **21**, and can magnetically attract the first magnetic element **12**. Then, in operation of the engaging module of the present invention, the multiple positioning protrusions **213** can enter or leave the positioning channels **133** via the guiding notches **131** and the extending channels **132** of the corresponding matching structures **13**.

(11) Furthermore, with reference to FIGS. **1** to **3**, the linking end **214** is formed on the other one of the two ends of the second casing **21**, and the second through hole **215** is formed through the linking end **214** of the second casing **21**.

(12) In addition, with reference to FIGS. **2** and **3**, a curvature of the extending channel **132** of each

matching structure **13** is a brachistochrone curve, and this enables a corresponding positioning protrusion **213** to move into the positioning channel **133** quickly via the extending channel **132**.
(13) The engaging module of the present invention can be used to connect two opposite ends of an object, the connecting end **113** of the male engaging element **10** and the linking end **214** of the female engaging element **20** are respectively connected to the two ends of the object, and the two ends of the object are connected to each other to form a loop by connecting the male engaging element **10** with the female engaging element **20**.
(14) With reference to FIGS. **4** and **5**, when the engaging module of the present invention is used by a user, the user moves the mounting end **211** of the female engaging element **20** to the assembling end **11** of the male engaging element **10**, and the first magnetic element **12** in the assembling end **111** of the male engaging element **10** is magnetically attracted to the second magnetic element **22** in the mounting hole **212** of the female engaging element **20**. Then the assembling end **111** of the male engaging element **10** extends into the mounting hole **212** of the female engaging element **20**, and each one of the multiple positioning protrusions **213** moves at the guiding notch **131** of a corresponding matching structure **13** on the external surface of the first casing **11** of the male engaging element **10**. In the process of mutual magnetic attraction between the first magnetic element **12** and the second magnetic element **22**, the male engaging element **10** and the female engaging element **20** are relatively moved, each one of the multiple positioning protrusions **213** of the second casing **21** of the female engaging element **20** moves in the extending channel **132** via the corresponding guiding notch **131** and further moves into the positioning area **134** of the positioning channel **133** via the corresponding extending channel **132**. Therefore, the user can quickly assemble the male engaging element **10** and the female engaging element **20** by magnetic attraction without alignment, and this can effectively improve the convenience of assembly.
(15) With reference to FIGS. **5** and **6**, when the male engaging element **10** and the female engaging element **20** of the engaging module are assembled, the magnetic attraction between the first magnetic element **12** and the second magnetic element **22** can prevent the male element separating from the female engaging element **20**, and each one of the multiple positioning protrusions **213** can be limited in the corresponding positioning channel **133**. When the male engaging element **10** and the female engaging element **20** are pulled by an external force, the positioning protrusion **213** can only move from the positioning area **134** to the limiting area **135** under the guidance of the positioning channel **133** without moving out from the extending channel **132**. After the external force is removed from the first engaging element **10** and the second engaging element **20**, the magnetic force between the first magnetic element **12** and the second magnetic element **22** will make the positioning protrusion **213** return to the positioning area **134** of the corresponding positioning channel **133**. Therefore, the male engaging element **10** and the female engaging element **20** will not be completely separated when pulled by an external force and this can effectively improve the stability of assembly.
(16) In addition, with reference to FIGS. **5** and **6**, the limiting protrusion **136** of each one of the multiple matching structures **13** can prevent the corresponding positioning protrusion **213** leaving from the corresponding positioning channel **113** when the male engaging element **10** and the female engaging element **20** are pulled by an external force, the non-return protrusion **137** can limit the corresponding positioning protrusion **213** in the positioning area **134** of the corresponding positioning channel **133**, and this can prevent the corresponding positioning protrusion **213** sliding out of the extension channel **132** when the male engaging element **10** and the female engaging element **20** of the engaging module are relatively rotated by a torsion force of the object itself.
(17) In summary, the male engaging element **10** and the female engaging element **20** of the engaging module of the present invention can be magnetically attracted to each other by the first magnetic element **12** and the second magnetic element **22**, and can avoid separating from each other when pulled by an external force by the structural relationship between each positioning

protrusion **213** of the second casing **21** of the female engaging element **20** and a corresponding one of the multiple matching structures **13** of the male engaging element **10**. The corresponding positioning protrusion **213** can be buckled into the positioning channel **133** of the corresponding matching structure **13** when the first magnetic element **12** and the second magnetic element **22** are attracted to each other, and will not leave away from the corresponding positioning channel **133** when pulled by an external force, and this can simultaneously improve the convenience of assembly and the stability after assembly.
(18) Even though numerous characteristics and advantages of the present invention have been set forth in the foregoing description, together with details of the structure and features of the invention, the disclosure is illustrative only. Changes may be made in the details, especially in matters of shape, size, and arrangement of parts within the principles of the invention to the full extent indicated by the broad general meaning of the terms in which the appended claims are expressed.

Claims

1. An engaging module comprising a male engaging element and a female engaging element, and the male engaging element assembled with the female engaging element along an assembling axis, wherein the male engaging element has a first casing having two ends; an assembling end formed on one of the two ends of the first casing; and an assembling hole formed in the assembling end of the first casing; multiple matching structures disposed around an external surface of the first casing, and each one of the multiple matching structures having a guiding notch being V-shaped and formed on the external surface of the first casing adjacent to the assembling end of the first casing; an extending channel formed on the external surface of the first casing, communicating with the guiding notch, and obliquely extended away from the assembling end of the first casing; a positioning channel formed on the external surface of the first casing, connected to and communicating with the extending channel opposite to the guiding notch, and having two ends; a positioning area formed on one of the two ends of the positioning channel that obliquely extends away from the assembling end of the first casing; and a limiting area formed on the other one of the two ends of the positioning channel that obliquely extends toward the assembling end of the first casing, and communicating with the positioning area; and wherein the guiding notches of the multiple matching structures are connected to and communicate with each other; a first magnetic element disposed in the assembling hole of the first casing; and the female engaging element has a second casing having two ends; a mounting end formed on one of the two ends of the second casing for the assembling end of the first casing to extend into the second casing; a mounting hole formed in the mounting end of the second casing; and multiple positioning protrusions formed on and protruding from an internal surface of the second casing and respectively corresponding to the multiple matching structures of the male engaging element; and a second magnetic element disposed in the mounting hole of the second casing, and selectively and magnetically attracted to the first magnetic element; wherein the multiple positioning protrusions respectively enter or leave the positioning channels via the guiding notches and the extending channels of the corresponding matching structures.
2. The engaging module as claimed in claim 1, wherein the first casing has a connecting end formed on the other one of the two ends of the first casing; and a first through hole formed through the connecting end of the first casing; and the second casing has a linking end formed on the other one of the two ends of the second casing; and a second through hole formed through the linking end of the second casing.
3. The engaging module as claimed in claim 2, wherein each one of the multiple matching structures has a limiting protrusion formed on and protruding from the external surface of the first casing between the limiting area and the extending channel of the matching structure.

4. The engaging module as claimed in claim 3, wherein each one of the multiple matching structures has a non-return protrusion formed on and protruding from the external surface of the first casing between the positioning area and the extending channel of the matching structure.
5. The engaging module as claimed in claim 4, wherein an imaginary connecting line is formed between the limiting protrusion and the non-return protrusion of each matching structure and is non-parallel to the assembling axis.
6. The engaging module as claimed in claim 2, wherein each one of the multiple matching structures has a non-return protrusion formed on and protruding from the external surface of the first casing between the positioning area and the extending channel of the matching structure.
7. The engaging module as claimed in claim 1, wherein each one of the multiple matching structures has a limiting protrusion formed on and protruding from the external surface of the first casing between the limiting area and the extending channel of the matching structure.
8. The engaging module as claimed in claim 7, wherein each one of the multiple matching structures has a non-return protrusion formed on and protruding from the external surface of the first casing between the positioning area and the extending channel of the matching structure.
9. The engaging module as claimed in claim 8, wherein an imaginary connecting line is formed between the limiting protrusion and the non-return protrusion of each matching structure and is non-parallel to the assembling axis.
10. The engaging module as claimed in claim 1, wherein each one of the multiple matching structures has a non-return protrusion formed on and protruding from the external surface of the first casing between the positioning area and the extending channel of the matching structure.